



## **Lab Task(s) Report – Lab 12**

**Name:**

**Muhammad Abu Bakar**

**NU ID:**

**24K-0826**

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**Syllabus:**

**Database Systems Lab**

**Syllabus Code: CL-2005**

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**Instructor:**

**Sir Osama Tahir**

Q1)

**MongoDB Shell Query Text:**

```
use SchoolDB
```

**Screenshot:**

```
> use schoolDB
< switched to db schoolDB
> show dbs
```

Q2)

**MongoDB Shell Query Text:**

```
db.createCollection("Students")
db.createCollection("Courses")
```

**Screenshot:**

```
> db.createCollection("Students")
db.createCollection("Courses")
< { ok: 1 }
schoolDB>
```

Q3)

**MongoDB Shell Query Text:**

```
db.Students.insertMany([
  { "_id": 1, "name": "Alice", "age": 20, "scores": { "math": 85, "science": 90 } },
  { "_id": 2, "name": "Bob", "age": 22, "scores": { "math": 78, "science": 82 } },
  { "_id": 3, "name": "Charlie", "age": 21, "scores": { "math": 92, "science": 88 } },
  { "_id": 4, "name": "Daisy", "age": 23, "scores": { "math": 68, "science": 74 } }
])
```

**Screenshot:**

```

> db.Students.insertMany([
  { "_id": 1, "name": "Alice", "age": 20, "scores": { "math": 85, "science": 90 } },
  { "_id": 2, "name": "Bob", "age": 22, "scores": { "math": 78, "science": 82 } },
  { "_id": 3, "name": "Charlie", "age": 21, "scores": { "math": 92, "science": 88 } },
  { "_id": 4, "name": "Daisy", "age": 23, "scores": { "math": 68, "science": 74 } }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': 1,
    '1': 2,
    '2': 3,
    '3': 4
  }
}
schoolDB>

```

Q4)

**MongoDB Shell Query Text:**

```

db.Courses.insertMany([
  { "_id": 101, "courseName": "Mathematics", "instructor": "Dr. Smith", "studentsEnrolled": [1, 2, 3] },
  { "_id": 102, "courseName": "Science", "instructor": "Dr. Adams", "studentsEnrolled": [2, 3, 4] }
])

```

**Screenshot:**

```

> db.Courses.insertMany([
  { "_id": 101, "courseName": "Mathematics", "instructor": "Dr. Smith", "studentsEnrolled": [1, 2, 3] },
  { "_id": 102, "courseName": "Science", "instructor": "Dr. Adams", "studentsEnrolled": [2, 3, 4] }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': 101,
    '1': 102
  }
}
schoolDB>

```

### Q5 – Part 1)

#### MongoDB Shell Query Text:

```
db.Students.findOne( { "scores.math": { $gte: 85 }, "age": { $lt: 22 } } )
```

#### Screenshot:

```
> db.Students.findOne( { "scores.math": { $gte: 85 }, "age": { $lt: 22 } } )
< {
  _id: 1,
  name: 'Alice',
  age: 20,
  scores: {
    math: 85,
    science: 90
  }
}
schoolDB>
```

### Q5 – Part 2)

#### MongoDB Shell Query Text:

```
db.Courses.findOne( { "studentsEnrolled": 3, "instructor": "Dr. Adams" } )
```

#### Screenshot:

```
> db.Courses.findOne( { "studentsEnrolled": 3, "instructor": "Dr. Adams" } )
< {
  _id: 102,
  courseName: 'Science',
  instructor: 'Dr. Adams',
  studentsEnrolled: [
    2,
    3,
    4
  ]
}
schoolDB> |
```

## Q6 – Part 1)

### MongoDB Shell Query Text:

```
db.Students.find({ "scores.math": { $gte: 80 }, "scores.science": { $lt: 90 } })
```

### Screenshot:

```
> db.Students.find({ "scores.math": { $gte: 80 }, "scores.science": { $lt: 90 } })
< {
  _id: 3,
  name: 'Charlie',
  age: 21,
  scores: {
    math: 92,
    science: 88
  }
}
schoolDB>
```

## Q6 – Part 2)

### MongoDB Shell Query Text:

```
db.Students.find({ $or: [ { "age": { $lt: 23 } }, { "scores.math": { $gte: 85 } } ] })
```

**Screenshot:**

```

> db.Students.find({ $or: [ { "age": { $lt: 23 } }, { "scores.math": { $gte: 85 } } ] })
< {
  _id: 1,
  name: 'Alice',
  age: 20,
  scores: {
    math: 85,
    science: 90
  }
}
{
  _id: 2,
  name: 'Bob',
  age: 22,
  scores: {
    math: 78,
    science: 82
  }
}
{
  _id: 3,
  name: 'Charlie',
  age: 21,
  scores: {
    math: 92,
    science: 88
  }
}
schoolDB> |

```

**Q6 – Part 3)****MongoDB Shell Query Text:**

```

db.Students.find({
  "scores.science": { $gte: 80 },
  $or: [ { "scores.math": { $lt: 75 } }, { "age": { $gt: 22 } } ]
})

```

**Screenshot:**

```
> db.Students.find({
  "scores.science": { $gte: 80 },
  $or: [ { "scores.math": { $lt: 75 } }, { "age": { $gt: 22 } } ]
})
<
schoolDB>
```

Q7)

**MongoDB Shell Query Text:**

```
db.Students.updateOne(
  { "name": "Bob", "scores.math": { $gte: 75 } },
  { $inc: { "scores.science": 1 } }
)
```

**Screenshot:**

```
> db.Students.updateOne(
  { "name": "Bob", "scores.math": { $gte: 75 } },
  { $inc: { "scores.science": 1 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
schoolDB>
```

Q8)

**MongoDB Shell Query Text:**

```
db.Students.updateMany(
  { "scores.science": { $lt: 80 }, "age": { $gt: 22 } },
  { $inc: { "scores.math": 5 } }
)
```

**Screenshot:**

```
> db.Students.updateMany(
  { "scores.science": { $lt: 80 }, "age": { $gt: 22 } },
  { $inc: { "scores.math": 5 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
schoolDB>
```

Q9)

**MongoDB Shell Query Text:**

```
db.Students.deleteOne({ "name": "Daisy", "scores.science": { $lt: 80 } })
```

**Screenshot:**

```
> db.Students.deleteOne({ "name": "Daisy", "scores.science": { $lt: 80 } })
< {
  acknowledged: true,
  deletedCount: 1
}
schoolDB> |
```

Q10)

**MongoDB Shell Query Text:**

```
db.Courses.deleteMany({
  $or: [
    { "studentsEnrolled": 2 },
    { "instructor": "Dr. Smith" }
  ]
})
```



**Screenshot:**

```

> db.Courses.deleteMany({
  $or: [
    { "studentsEnrolled": 2 },
    { "instructor": "Dr. Smith" }
  ]
})
< {
  acknowledged: true,
  deletedCount: 2
}
schoolDB>

```

Q11)

**MongoDB Shell Query Text:**

```
db.Students.drop()
```

**Screenshot:**

```

> db.Students.drop()
< true
schoolDB>

```

Q12)

**MongoDB Shell Query Text:**

```
db.Courses.drop()
```

**Screenshot:**

```

> db.Courses.drop()
< true
schoolDB>

```

Q13)

**MongoDB Shell Query Text:**

```
db.dropDatabase()
```

**Screenshot:**

```
> db.dropDatabase()  
< { ok: 1, dropped: 'schoolDB' }  
schoolDB>
```